

Day 1 - Print Numbers from 1 to N

Question

Ek number N diya gaya hai.

1 se N tak ke numbers print karo.

Example

Input:

5

Output:

1
2
3
4
5

Approach 1 - Using for Loop

Logic

1 se start karo.

Har iteration me number print karo.

N tak repeat karo.

Dry Run

N = 5

```
i = 1 → print 1  
i = 2 → print 2  
i = 3 → print 3  
i = 4 → print 4  
i = 5 → print 5
```

Time Complexity

$O(N)$

Space Complexity

$O(1)$

Java Code

```
public class Day1 {  
    public static void main(String[] args) {  
        int n = 5;  
  
        for(int i = 1; i <= n; i++) {  
            System.out.println(i);  
        }  
    }  
}
```

Approach 2 - Using While Loop

Logic

Jab tak number N se chhota ya equal hai tab tak print karte raho.

Time Complexity

$O(N)$

Space Complexity

$O(1)$

Java Code

```
public class Day1 {  
    public static void main(String[] args) {  
        int n = 5;  
        int i = 1;  
  
        while(i <= n) {  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```

Approach 3 - Using Recursion

Logic

Function ko baar-baar call karo jab tak N tak na pahunch jao.

Time Complexity

$O(N)$

Space Complexity

$O(N)$

(Kyunki recursion call stack use karta hai)

Java Code

```
public class Day1 {  
    static void printNumbers(int i, int n) {  
        if(i > n)  
            return;  
  
        System.out.println(i);  
  
        printNumbers(i + 1, n);  
    }  
  
    public static void main(String[] args) {  
        int n = 5;  
  
        printNumbers(1, n);  
    }  
}
```

Interview Discussion

Interviewer puch sakta hai:

Q1. for loop aur while loop me difference?

Answer:

- for loop → iterations pata ho
 - while loop → iterations pata na ho
-

Q2. Recursion ka disadvantage?

Answer:

- Extra stack memory use hoti hai
 - Stack Overflow aa sakta hai
-

Q3. Best Approach?

Answer:

For Loop

Time : $O(N)$
Space : $O(1)$

Ye sabse efficient hai.

Summary

Approach Time Space

For Loop $O(N)$ $O(1)$

While Loop $O(N)$ $O(1)$

Recursion $O(N)$ $O(N)$